The Phonetics of Cantonese

Background

Cantonese is a major Chinese variety in the Yue branch of Sino-Tibetan languages. It is primarily spoken in southern China, particularly in Guangdong Province, Hong Kong, and Macau. The language originated in and around Guangzhou (historically known as Canton City), hence its name. Cantonese has also spread globally through immigrant communities, with significant populations in Southeast Asia, North America, and Australia. According to Eberhard, Simons, and Fennig (2021), there are approximately 85 million native Cantonese speakers worldwide. While Cantonese does not have a standardized written form, it is often written using Standard Chinese characters, either for their semantic value or phonetic similarity to Cantonese pronunciations. This results in various writing conventions, including the use of some unique characters specific to Cantonese. Because it is spoken by many people across all regions, Cantonese also has many different dialects and pronounces the same Standard Chinese characters differently. Variations may even appear between small villages and towns within the Guangdong region.

The language consultant for this study is Vincent Pan, an 18-year-old first-year student at UCLA. He comes from Shenzhen, a city located close to Guangzhou and Hong Kong, and has been largely influenced by both of their cultures. Vincent speaks the Guangzhou dialect of Cantonese, which he acquired growing up in a Cantonese-speaking family. His formal education has been primarily conducted in Mandarin Chinese and English, and these are also the primary languages he uses to communicate with others. In his daily life, Vincent rarely initiates conversations in Cantonese but can still comfortably converse with Cantonese speakers. While he cannot read or write Cantonese specifically, he can read Standard Chinese (written Mandarin) and mentally translate it to spoken Cantonese when he needs to read them out.

The primary reference for this report is “Modern Cantonese Phonology” by Robert S. Bauer and Paul K. Benedict (2011). This comprehensive work provides an in-depth analysis of Cantonese phonology, including detailed examinations of consonants, vowels, and tones. The book also offers extensive examples of lexical items that differentiate sounds that are very similar to one another and discusses various allophones, making it a valuable resource for understanding the sound system of Cantonese. Bauer and Benedict’s work is particularly useful for its thorough documentation of phonological processes and its comparison of Cantonese with other Chinese varieties.

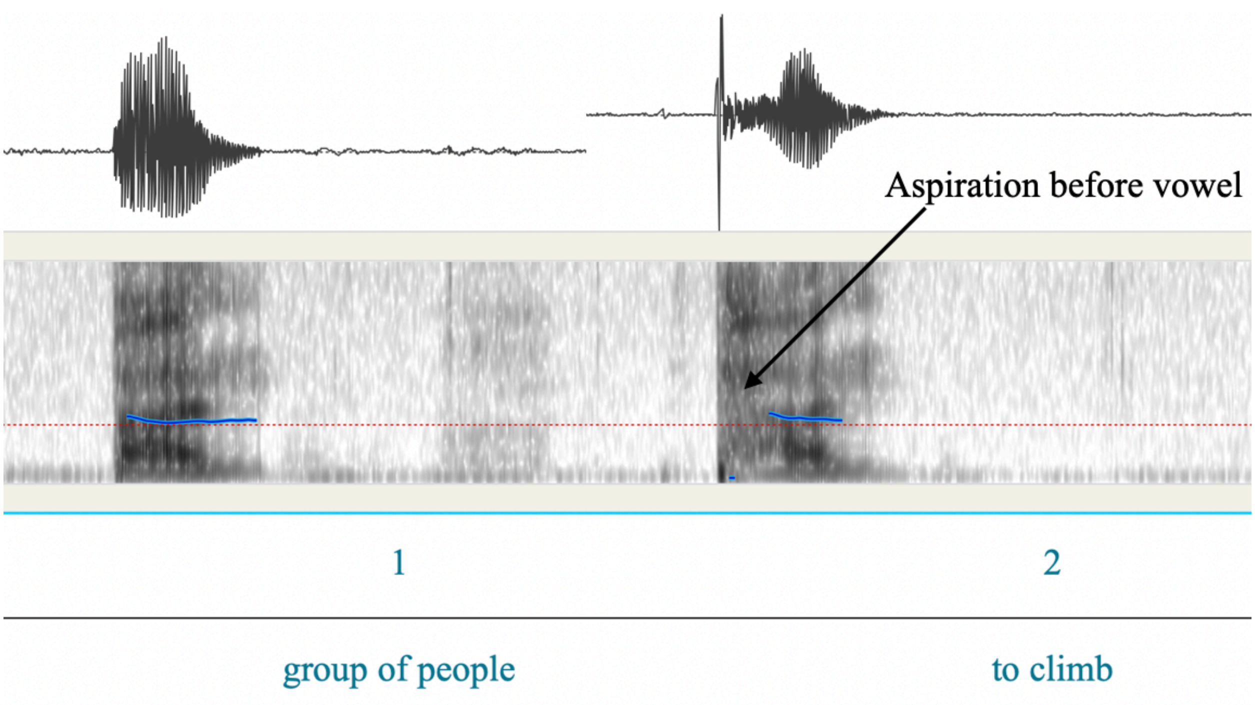
Consonants

There are 19 consonants in Cantonese as shown in the following chart. Voiceless sounds are on the left and voiced sounds are on the right.

	Labial	Dental/ Alveolar	Post-alveolar / Palatal	Velar	Labial-Velar	Glottal
Plosives	p	t		k	k^w	
Aspirated Plosives	p^h	t^h		k^h	k^{hw}	
Nasal	m	n		ŋ		
Fricative	f	s				h
Affricate		ts				
Aspirated Affricate		ts^h				
Approximant	w	l	j			

Plosives

Cantonese has eight voiceless plosives categorized into unaspirated plosives, aspirated plosives, unaspirated labialized plosives, and aspirated labialized plosives. The three unaspirated plosives are the bilabial stop /p/, the alveolar stop /t/, and the velar stop /k/. The other three aspirated plosives are all aspirated versions of the unaspirated plosives. The pair of labialized plosives are /k^w/ and /k^{hw}/, which only differ in aspiration. All eight plosives can appear in the word-initial position, but only the three unaspirated plosives can appear in the word-final position. In the recording, the speaker Vincent demonstrates the aspiration very clearly. Shown in Graph 1, the difference between aspiration in #1 /paːn˥/ and #2 /p^haːn˥/ is evident, corresponding to Bauer and Benedict's description and the expected pronunciation. When in word-final position, the unaspirated plosives become unreleased consonants, an allophonic variation that will be elaborated on in the Allophone section.



Graph 1: Comparison between unaspirated /p/ and aspirated /p^h/. Graph is edited to align two words with common pitch line. There's a clear aspiration period with #2.

Nasal

Cantonese has three nasal stops: voiced bilabial nasal /m/, voiced alveolar nasal /n/, and voiced velar nasal /ŋ/. All three nasal stops can occur in word-initial and final position. Bauer and Benedict discussed how various speakers exhibit different pronunciations for words with /ŋ/ and /n/ as word-initial consonants as many dialects of Cantonese substitute/n/ with /l/ or demonstrate initial-ŋ-dropping. In Vincent's production, the /n/ in #10 /naː˧/ is not substituted, and the /ŋ/ in #11 /ŋaː˩/ is present. Unlike the unaspirated plosives, which become unreleased consonants at word-final positions, the three nasals have no expected variants when occurring after a vowel at the end of the syllable. Another interesting feature of the nasal sounds in Cantonese is that the nasal consonants /m/ and /ŋ/ can occur as a complete independent syllable without being followed by vowels. An example of this is #49 /ŋ˩˧/. Even though in Bauer and Benedict's observation, younger generation of Cantonese speakers tend to substitute the syllabic /ŋ/ with [m] and none of their consulted speakers under 50 years old used /ŋ/, Vincent pronounces both #11 /ŋaː˩/ and #49 /ŋ˩˧/ with [ŋ]. Interestingly, when pronouncing #37 /taːm˥/, he first pronounced it as [taːm˥] and then pronounced it as [taːŋ˥], substituting /m/ with [ŋ] and demonstrating the opposite of Bauer and Benedict's findings. However, given that Vincent produces both #5 /kaːm˧/ and #6 /kʰaːm˧/ with [m], it is not a common nor general variation of the phoneme.

Fricatives

Cantonese has three voiceless fricatives: labio-dental /f/, alveolar /s/, and glottal /h/. They can only appear in word-initial position shown by the minimal triplet #12 /faː˥/, #13 /saː˥/, and

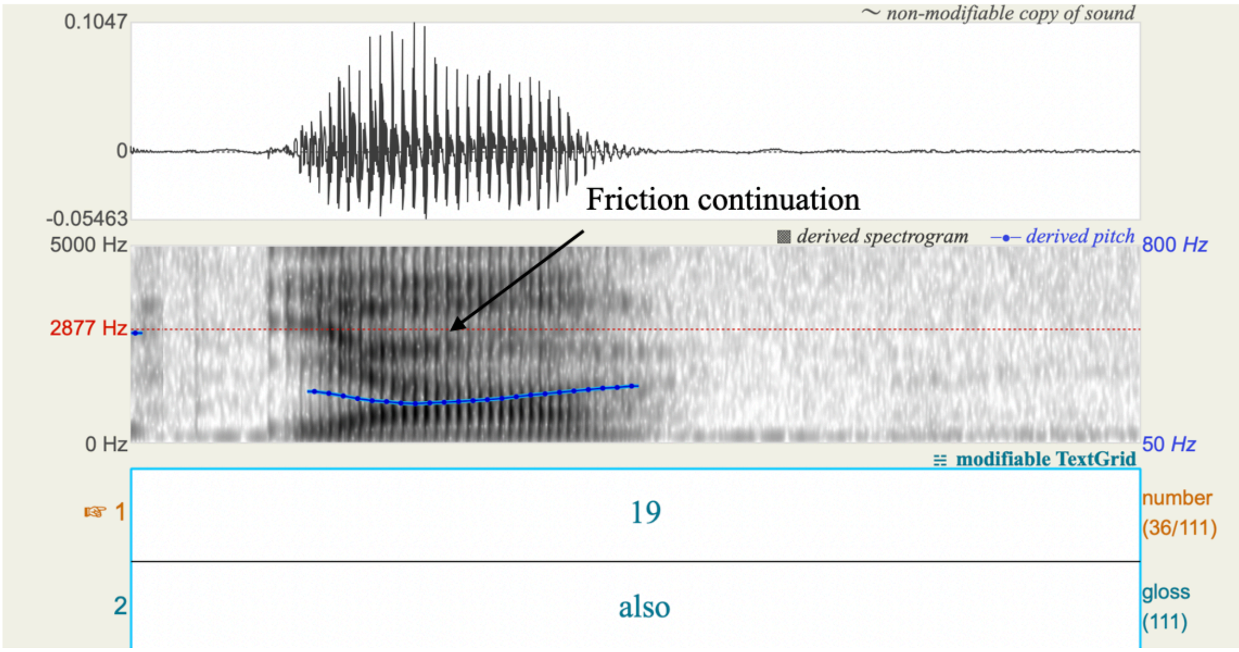
#16 /haː˥/. Vincent produces the three fricatives as expected. The alveolar fricative /s/ has a palatal allophone, which will be elaborated on in the Allophone section.

<u>Affricates</u>

Cantonese has a pair of affricates that differ only in aspiration: /ts/ and /tsʰ/ shown by the minimal pair #14 /tsaː˥/ and #15 /tsʰaː˥/. Vincent produces the two affricates as expected. Both affricates experience allophonic variation when followed by close front vowels /iː/ and /yː/, which will be elaborated on in the Allophone section.

<u>Approximants</u>

Cantonese has three voiced approximants: labial /w/, lateral /l/, and palatal /j/. They can appear as word-initial consonants. Bauer and Benedict observed the articulation in Cantonese for /w/ and /j/ being pronounced with noticeable friction when they appear in word-initial position, and that friction would continue into the following vowel sound. Despite not being obvious in audio, the Spectrogram image in Graph 2 shows that the /j/'s friction did continue into the vowels in #19 /jaː˩˧/. /w/ and /j/ can also be combined with vowels to form diphthongs at the end of the syllables, which will be discussed in the Vowels section.



Graph 2: Friction continuation of /j/ in #19

Vowels and Rimes

Cantonese has eleven vowel phonemes. However, only seven of them can appear as independent rimes: /iː/, /yː/, /uː/, /ɛː/, /œː/, /ɔː/, and /aː/. These seven vowels also have a longer duration in words, indicated by the colon ː, compared to the vowels /e/, /ɐ/, /o/, and /ɵ/. The eleven vowels are combined with nine final consonants/m/, /n/, /ŋ/, /p/, /t/, /k/, /w/, /j/, and /y/ to form the rimes of Cantonese. The rimes can be described in three categories: monophthongs or independent rimes, diphthongs, and combined rimes.

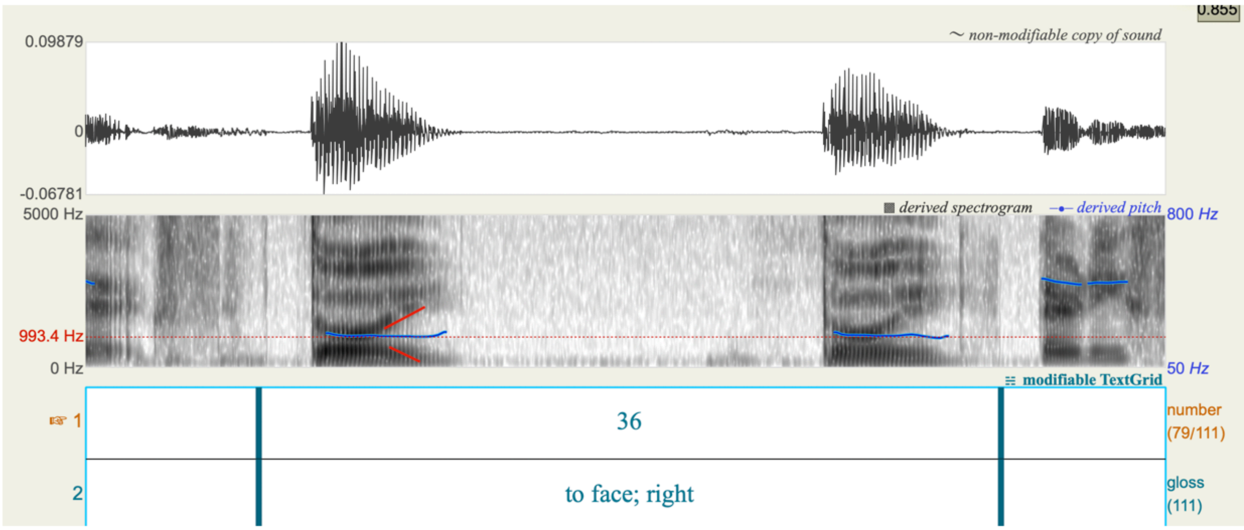
	Front	Central	Back
Close	i y		u
Close-Mid	e		o
Mid			ɵ
Open-Mid	ɛ œ	ɐ	ɔ
Open		a	

Monophthongs

Cantonese monophthongs are also known as independent rimes, which comprise the seven long vowels. Among the seven monophthongs, /iː/, /uː/, and /yː/ can not appear as independent syllables. /iː/ can appear before most initial consonants except for /k^{hw}/ and /ŋ/. /ɛː/ can appear before most initial consonants except for /k^{hw}/, /ŋ/, and /t^{h}/. /aː/ can appear before all initial consonants. /yː/ can only appear after the palatal approximant /j/ and the palatalized fricatives and affricates, which are the allophonic variation of /s/, /ts/, and /ts^{h}/. Bauer and Benedict also observed that /yː/ and /uː/ are in an almost perfect complementary distribution when they appear as monophthongs. Therefore,/uː/ can appear after all initial consonants except for those that could be followed by /yː/. /ɔː/ can appear after all initial consonants except for /k^{hw}/. /œː/ can appear after all initial consonants except for labially articulated consonants. Vincent produces all the monophthongs as expected except for #22 /tuː˥/, which he pronounced as [tow˥].

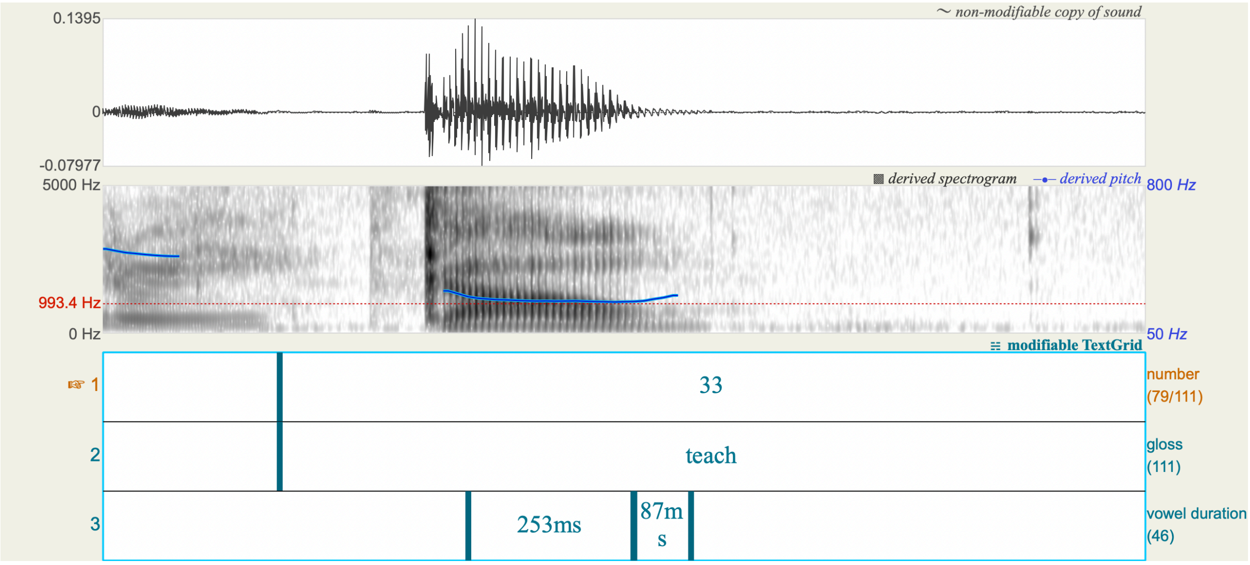
Diphthongs

There are eleven diphthongs in Cantonese. Cantonese diphthongs are constituted by a nuclear vowel followed by one of the approximants or the semivowels /w/, /j/, or /y/. Only one vowel, /ɵ/, can be combined with /y/ to form the diphthong /ɵy/. However, based on Bauer and Benedict's observation of the everyday speech of Cantonese speakers, /ɵy/ can also be analyzed as /ɵj/ with the /j/ rounding to [y]. Vincent's production for #36 /tɵy˧/ evidences this observation as he pronounced it as [tɵj˧], shown in Graph 3 by the change in vowel formant.

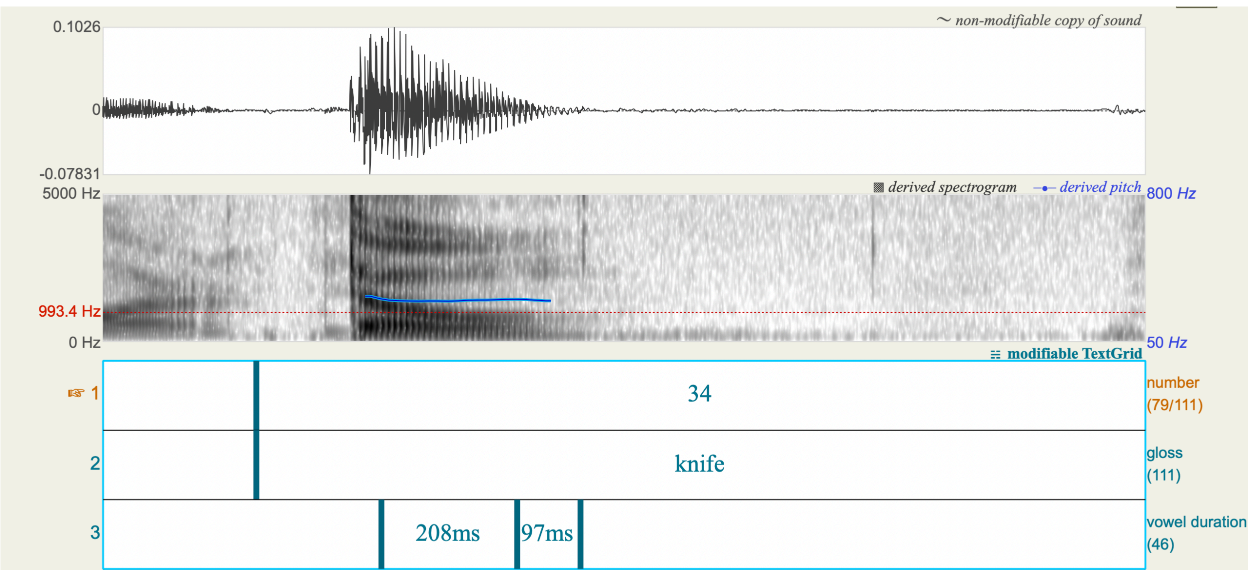


Graph 3: #36 /tɵy˧/ showing formant changes resembling [ɵy]

The other ten diphthongs are /ej/, /uːj/, /ɔːj/, /ɐj/, /aːj/, /ɐw/, /aːw/, /ow/, /iːw/, and /ɛw/. In reality, the diphthong /ɛw/ is rarely used and would only occur in colloquial Cantonese and English loanwords. The speaker Vincent also confirms that he does not know any word containing this diphthong. Bauer and Benedict also observed that the duration of long nuclear vowels is about three times longer than the semivowels, while the duration of short nuclear vowels is about two times longer. This phenomenon is also demonstrated in Vincent's production. When producing #33 /kaːw˧/, this Spectrogram image in Graph 4 exhibits the vowel duration differences of three times longer. In the following #34 /tow˥/ in Graph 5, the nuclear vowel [o] is also about two times longer than the semivowel [w], corresponding to Bauer and Benedict's observation.



Graph 4: #33 has a long nuclear vowel duration of 253ms and a semivowel duration of 87ms.

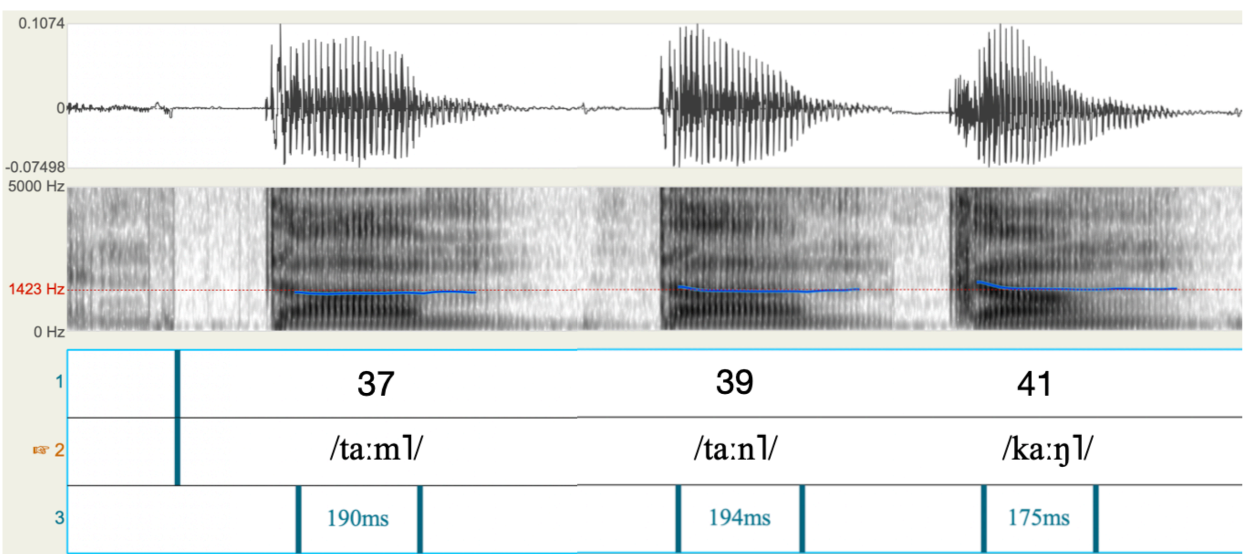


Graph 5: #34 has a short nuclear vowel duration of 208 ms and a semivowel duration of 97 ms.

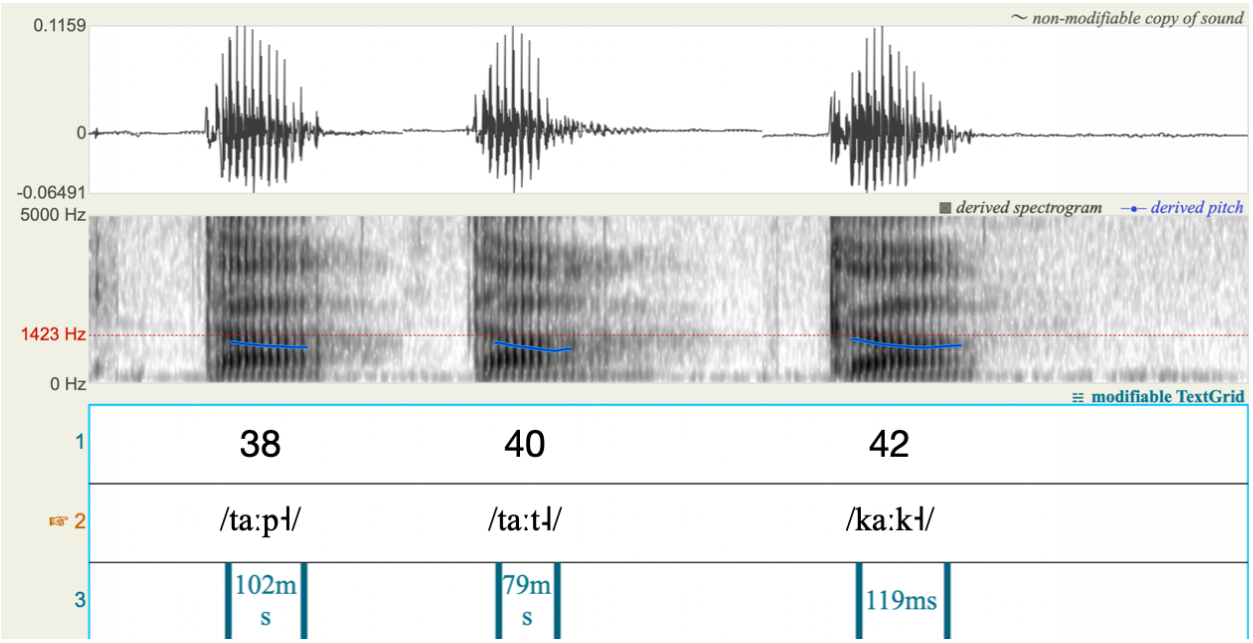
Combined Rimes

Cantonese rimes are the combinations of vowels and the six final consonants mentioned above. The stops are generally grouped into three pairs based on their places of articulation: /m/ and /p/, /n/ and /t/, and /ŋ/ and /k/. For the bilabial pair, only three long vowels,/iː/, /ɛː/, and /aː/, and a short vowel /ɐ/ can appear before them. For the alveolar pair, all seven long vowels and two short vowels, /ɐ/ and /ɵ/, can appear before them. For the velar pair, four long vowels /ɛ:/, /œ:/, /ɔ:/, and /a:/ and two short vowels /ɐ/ and /o/ can appear before them. Due to the large amount of rimes possible in Cantonese, this paper only looks at the six possible combinations /a:/ have with the six final consonants: #37 /ta:m˥/, #38 /ta:p˧/, #39 /ta:n˥/, #40 /ta:t˧/, #41 /ka:ŋ˥/ and #42 /ka:k˧/.

Bauer and Benedict noted that for the same vowel, rimes that end with nasal sounds generally have a longer vowel duration than those that end with plosives. My source evidences this claim, and the production of longer vowel durations for final nasal consonants is very consistent. Graph 6 shows the three words #37 /ta:m˥/, #39 /ta:n˥/, and #41 /ka:ŋ˥/ that end with nasal consonants. Graph 7 shows the three words #38 /ta:p˧/, #40 /ta:t˧/, and #42 /ka:k˧/ that ends with plosives which varied into unreleased consonants. The vowel durations in Graph 6 are significantly longer than those in Graph 7, corresponding to Bauer and Benedict's observation.



Graph 6

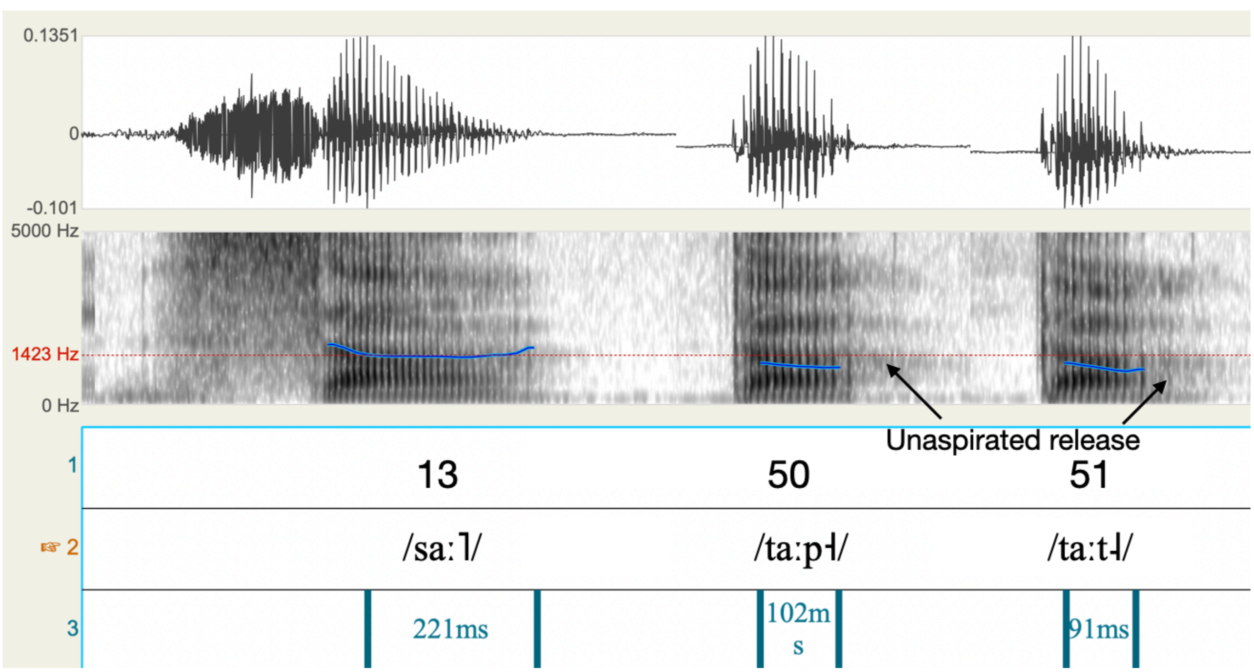


Graph 7

Allophones

Bauer and Benedict (2011) describe two types of allophonic variation in Cantonese in their book. The first type of allophonic variation happens when unaspirated plosives /p/, /t/, and /k/ occur in word-final position. The three plosives become unreleased consonants [p̚], [t̚], and [k̚]. Bauer and Benedict describe words exhibiting these three allophones as "dead syllables" as opposed to "open syllables," which are words that end with the vowels we discussed above. Tones in words that end with unreleased consonants are also given a special name despite showing the same tone contour as the six tones discussed above. They are called "entering tones" and only consist of the high-level tone, the mid-level tone, and the mid-low-level tone. The second type of allophonic variation is the palatalization of the alveolar fricatives and affricates /s/, /ts/, and /tsʰ/ when they occur before /i:/ and /y:/. Based on different dialects, they could either be palatalized into [ʃ], [tʃ], and [tʃʰ] or [ɕ], [tɕ], and [tɕʰ]. In some dialects, the /s/ would be palatalized regardless of the vowels that follow.

Vincent demonstrates both allophonic variations in his production. He consistently pronounced the unaspirated plosives as unreleased consonants when they occur at the end of the word. He pronounced #50 /ta:p˧/ as [ta:p̚˧] and #51 /ta:t˧/ as [ta:t̚˧]. However, despite not releasing the air at the end of the world, based on the Spectrogram image below, there are still differences between words that end with vowels and those that end with unreleased consonants. Shown in Graph 8, compared to a word #13 /sa:˥/ that contains the same vowel /a:/, #50 /ta:p˧/ and #51 /ta:t˧/ have shorter vowel duration and despite lacking the normal release burst or aspiration, the consonants are held and then released without accompanying aspirations.



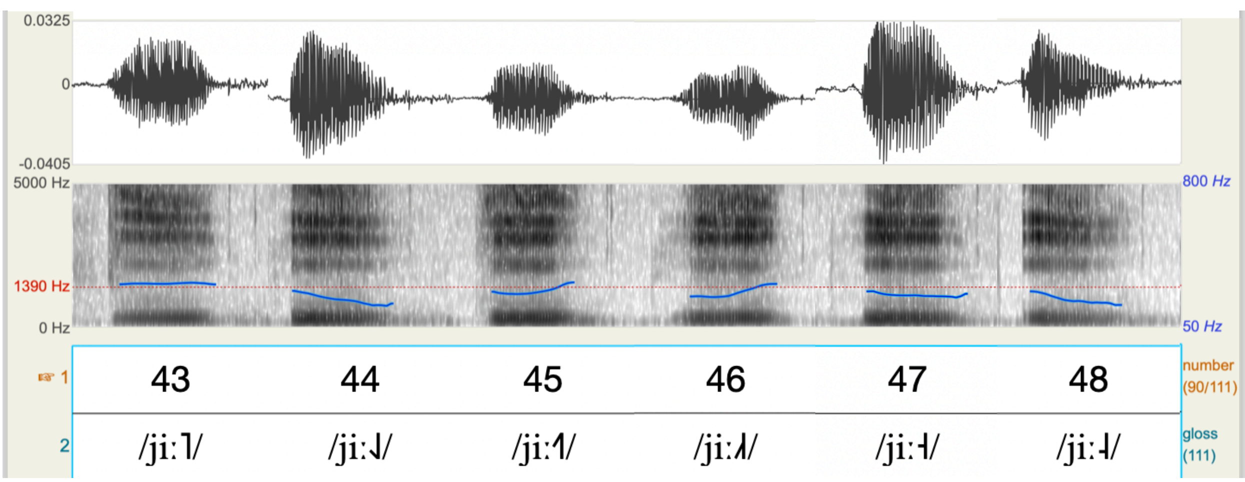
Graph 8: Comparison of vowel length and evidence of unaspirated release of the unreleased plosives

The second type of allophonic variation is demonstrated inconsistently in Vincent's production. He exhibits the palatalization process in #52 /syː˥/and #54 /tsʰyː˥/, which he pronounced as #52 [ɕyː˥] and #54 [ɕyː˧]. But for #20 /siː˥/ and #53 /tsiː˥/, despite also being followed by /iː/, there is no obvious palatalization. A plausible explanation for the difference in palatalization is that despite /iː/ and /yː/ both being close front vowels, /yː/'s roundedness requires the lip rounding and raising the tongue toward the hard palate, which subsequently makes palatalization more obvious.

Suprasegmentals

Tone is the most notable suprasegmental feature of Cantonese. The tone of each word is a crucial part of its pronunciation, as a change in tone would imply a different word despite the consonants and the vowels remaining the same. Cantonese has six tones: high level, mid-low falling, high rising, mid-low rising, mid-level, and mid-level. For native Cantonese speakers, the six tones are grouped into two triplets, categorized by dark (Yin) or bright (Yang). In each group, there is a level tone, a rising tone, and a going tone. The high-level tone /˥/ is also known as the dark-level tone and can remain at its high level throughout or experience a slight fall toward its end. The mid-low falling tone /˨˩/ is also known as the bright level tone. It starts at mid-low level and falls to a lower point. The high rising tone /˧˥/, or the dark rising tone, rises from mid-low level to high or first experiences a fall from mid to mid-low before rising to high. Bauer and Benedict noted that the high-rising tone normally starts at the same level as the mid-low rising tone but rises to a higher point. The mid-low rising tone /˨˧/, or the bright rising tone, starts at the mid-low level and rises to the mid-point. The mid-level tone /˧/, or the dark-going tone, is similar to the high-level tone but instead occupies the mid-level. The low-level tone /˨/, or the bright going tone, starts at the mid-low point and may experience slight falling toward its end while not completely falling to the low point.

Vincent's production is fairly consistent with Bauer and Benedict's description. By comparing the pitch of the six words #43 /ji:˥/, #44 /ji:˨˩/, #45 /ji:˧˥/, #46 /ji:˨˧/, #47 /ji:˧/, and #48 /ji:˨/. We can differentiate between the six tones despite them sounding similar to one another. In Graph 9, the six words are placed together to give a clearer image of their pitch differences. With 1390 Hz as the aligning pitch value, the high-level tone #43 has a higher pitch than other tones. The mid-low level and low-level tone both start at approximately the same pitch, but there's a more significant fall in #48, while #47 majorly remains level. #45 and #46, which are the two rising tones, are harder to distinguish as they both show a rising trend and end in a similar pitch. The only visible difference is that #45 starts at a higher pitch than #46.



Graph 9: Tone Pitch Comparison. Graph is edited to include all six tones aligned by the common pitch line of 1390 Hz.

Word List

Consonants						
#	Phoneme	Phonemic	Orthography	Gloss	Phonetic 1	Phonetic 2
1	/p/	/pa:n˥/	班	'group of people'	[pa:n˥]	[pa:n˥]
2	/p^h/	/p^{h}a:n˥/	攀	'to climb'	[p^{h}a:n˥]	[p^{h}a:n˥]
3	/t/	/ta:n˥/	单	'single'	[ta:n˥]	[ta:n˥]
4	/t^h/	/t^{h}a:n˥/	摊	'to open and spread out'	[t^{h}a:n˥]	[t^{h}a:n˥]
5	/k/	/ka:m˧˥/	噉	'in this way; in that case'	[ka:m˧˥]	[ka:m˧˥]
6	/k^h/	/k^{h}a:m˧˥/	冚	'to cover'	[k^{h}a:m˧˥]	[k^{h}a:m˧˥]
7	/k^w/	/k^{w}a:˥/	瓜	'melon'	[k^{w}a:˥]	[k^{w}a:˥]
8	/k^{hw}/	/k^{hw}a:j˥/	规	'rule'	[k^{hw}ej˥]	[k^{hw}ej˥]
9	/m/	/ma:˥/	妈	'mother'	[ma:˥]	[ma:˥]
10	/n/	/na:˧˥/	乸	'female'	[na:˧˥]	[na:˧˥]
11	/ŋ/	/ŋa:˨˩/	牙	'tooth'	[ŋa:˨˩]	[ŋa:˨˩]
12	/f/	/fa:˥/	花	'flower'	[fa:˥]	[fa:˥]
13	/s/	/sa:˥/	沙	'sand'	[sa:˥]	[sa:˥]
14	/ts/	/tsa:˥/	揸	'to grab'	[tsa:˥]	[tsa:˥]
15	/ts^h/	/ts^{h}a:˥/	差	'difference; bad'	[ts^{h}a:˥]	[ts^{h}a:˥]
16	/h/	/ha:˥/	虾	'shrimp'	[ha:˥]	[ha:˥]
17	/w/	/wa:˨/	话	'to say'	[wa:˨]	[wa:˨]
18	/l/	/la:˥/	啦	(sentence final part)	[la:˥]	[la:˥]
19	/j/	/ja:˩˧/	也	'also'	[ja:˩˧]	[ja:˩˧]
Vowels, Diphthongs, and Rimes						
20	/i:/	/si:˥/	丝	'silk'	[si:˥]	[si:˥]

21	/yː/	/syː˥/	书	'book'	[ʃyː˥]	[ʃyː˥]
22	/uː/	/tuː˥/	都	'all'	[tow˥]	[tow˥]
23	/ɛː/	/tɛː˥/	爹	'father'	[tɛː˥]	[tɛː˥]
24	/œː/	/hœː˥/	靴	'boots'	[hœ˥]	[hœ˥]
25	/ɔː/	/tɔː˥/	多	'much, many'	[tɔː˥]	[tɔː˥]
26	/aː/	/saː˥/	沙	'sand'	[saː˥]	[saː˥]
27	/ej/	/tej˨/	地	'earth'	[tej˨]	[tej˨]
28	/uːj/	/muːj˥/	每	'every'	[mɔːj˧˥]	[mɔːj˧˥]
29	/ɔːj/	/kɔːj˨/	盖	'to cover'	[kɔːj˨]	[kɔːj˨]
30	/ɐj/	/bɐj˧˥/	闭	'to close’	[bɐj˨]	[bɐj˨]
31	/aːj/	/taːj˨/	大	'big'	[taːj˨]	[taːj˨]
32	/ɐw/	/kɐw˧/	救	'rescue'	[kow˧]	[kow˧]
33	/aːw/	/kaːw˧/	教	'teach'	[kaːw˧]	[kaːw˧]
34	/ow/	/tow˥/	刀	'knife'	[tow˥]	[tow˥]
35	/iːw/	/tiːw˥/	丢	'lose'	[tiːw˥]	[tiːw˥]
36	/ɵy/	/tɵy˧/	对	'to face; right'	[tɵj˧]	[tɵj˧]
37	/aːm/	/taːm˥/	担	'carry on shoulder pole'	[taːm˥]	[taːŋ˥]
38	/aːp/	/taːp˧/	搭	'board (vehicle)'	[taːp̚˧]	[taːp̚˧]
39	/aːn/	/taːn˥/	单	'single'	[taːn˥]	[taːn˥]
40	/aːt/	/taːt˨/	达	'to reach'	[taːt̚˨]	[taːt̚˨]
41	/aːŋ/	/kaːŋ˥/	耕	'to plow'	[kaːŋ˥]	[kaːŋ˥]
42	/aːk/	/kaːk˧/	格	'line'	[kaːk̚˧]	[kaːk̚˧]
Tones						
#	Tone	Phonemic	Orthography	Gloss	Phonetic 1	Phonetic 2

43	high level 阴平	/ji:˥/	衣	'clothes'	[ji:˥]	[ji:˥]
44	mid-low falling 阳平	/ji:˨˩/	疑	'suspicious'	[ji:˨˩]	[ji:˨˩]
45	high rising 阴上	/ji:˧˥/	椅	'chair'	[ji:˧˥]	[ji:˧˥]
46	mid-low rising 阳上	/ji:˨˧/	耳	'ear'	[ji:˨˧]	[ji:˨˧]
47	mid level 阴去	/ji:˧/	意	'idea'	[ji:˧]	[ji:˧]
48	mid-low level 阳去	/ji:˨/	二	'two'	[ji:˨]	[ji:˨]
Interesting Phonemes						
49	/ŋ/	/ŋ˨˧/	五	'five'	[ŋ˨˧]	[ŋ˨˧]
Allophones						
#	Phonemic	Allophone	Orthography	Gloss	Phonetic 1	Phonetic 2
When unaspirated stops occured at the end of the sentence, they become unreleased consontants.						
50	/ta:p˧/	[ta:p̚˧]	搭	'board (vehicle)'	[ta:p̚˧]	[ta:p̚˧]
51	/ta:t˨/	[ta:t̚˨]	达	'to reach'	[ta:t̚˨]	[ta:t̚˨]
/s/ /ts/ and /tsh/ become palatalized /ʃ/ /tʃ/ /tʃh/ when before						
52	/sy:˥/	[sy:˥] [ʃy:˥] [ɕy:˥]	书	'book'	[ɕy:˥]	[ɕy:˥]
53	/tsi:˥/	[tsi:˥] [tʃi:˥] [tɕi:˥]	字	‘character'	[tsi:˨]	[tsi:˨]

54	/tsʰyː˥/	[tsʰyː˥] [tʃʰyː˥] [tɕʰyː˥]	处	'location'	[ɕyː˧]	[ɕyː˧]

Sentences

55. 我想喝茶。 “I want to drink tea.” /ŋɔː˨˩˧ sœːŋ˥˧ hɔːt˧ tsʰaː˩/
[ŋɔː˨˩˧ sœːŋ˥˧ jɐm˥˧ tsʰaː˩] [ŋɔː˨˩˧ sœːŋ˥˧ jɐm˥˧ tsʰaː˩]

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